

AI Usage Summary Report

Project: Incident Management Agent – AI-Assisted Root Cause Analysis Platform

1. What AI Helped With

Artificial Intelligence was used as an engineering support and productivity tool throughout the complete software development lifecycle of the Incident Management Agent project. AI served as a technical assistant to accelerate analysis, planning, design, implementation support, testing preparation, and documentation activities while all final engineering decisions remained under the control of the development team.

Requirements Analysis & Planning

- Analyzed project requirements and objectives.
- Identified key features, roles, and project scope.
- Assisted in roadmap creation and risk assessment.

System Architecture & Design

- Assisted in designing frontend, backend, database, and API architecture.
- Planned authentication, AI components, and deployment strategy.

AI & Machine Learning Design

- Designed Severity Prediction and Category Classification workflows.
- Planned FAISS-based Similar Incident Retrieval and AI-powered RCA generation.
- Assisted with synthetic dataset creation and feature engineering.

Development Support

- Guided database design, API development, authentication, and RBAC.
- Assisted with incident management, reporting, and AI integration.

Testing & Validation

- Generated sample incidents, logs, datasets, RCA examples, and demo scenarios for testing and validation.

All AI-generated recommendations were reviewed, validated, modified, and implemented by the development team before inclusion in the final system.

2. What AI Got Wrong

Although AI significantly accelerated the project, several recommendations required human review and refinement before implementation.

Scope Expansion

Some suggestions introduced enterprise-scale features that exceeded the intended project scope, resulting in unnecessary complexity that required simplification.

Architectural Overengineering

Certain recommendations assumed:

- Large-scale production environments.
- High-volume datasets.
- Enterprise infrastructure.
- Distributed deployment architectures.

These recommendations were adjusted to better fit project requirements and available resources.

Workflow Complexity

AI occasionally proposed workflows that:

- Increased development effort.
- Added unnecessary user interactions.
- Reduced usability.

The team simplified these workflows to improve user experience and maintain a clean enterprise-focused interface.

These limitations reinforced the importance of engineering judgment and human validation throughout development.

3. Project Phases and Outcomes

Phase	Outcome
Phase 1 – Project Analysis & Development Planning	Requirement analysis, feature identification, architecture review, project roadmap creation, risk assessment, and UI/UX direction.
Phase 2 – System Architecture & Technical Design	Database design, API specifications, AI architecture planning, deployment strategy, security review, and technical documentation.
Phase 3 – Implementation & Development	Frontend implementation, backend services, authentication, incident management workflows, AI model integration, FAISS retrieval, RCA generation, reporting modules, and user management.
Phase 4 – Testing, Optimization & Deployment Readiness	Functional testing preparation, AI validation, performance optimization, deployment readiness review, documentation creation, and presentation preparation.

AI served as a technical accelerator throughout all phases, while implementation, testing, validation, and final decision-making remained the responsibility of the development team.

4. Best Prompts Used During Development

Prompt 1 – Architecture, Design & Planning Prompt

A comprehensive architecture prompt was used to instruct AI to act as a:

- Senior Software Architect
- Backend Engineer
- AI Engineer
- Database Designer
- Security Engineer
- UI/UX Designer

The prompt requested:

- Requirement analysis
- Business workflow analysis
- Architecture design
- Database design
- AI pipeline design
- Security review
- Deployment planning
- Risk identification
- Clarification questions before implementation

This prompt significantly accelerated project planning and architectural decision-making.

Example Excerpt

"Act as a Senior Software Architect, Senior Backend Engineer, Senior AI Engineer, Senior Database Designer, Senior Security Engineer, and Senior UI/UX Designer. Before generating implementation plans, analyze business objectives, user workflows, AI requirements, deployment requirements, and system architecture. Identify risks, assumptions, scalability concerns, and missing requirements."

Prompt 2 – Development Prompt

A master development prompt was used to guide implementation across all project phases.

The prompt instructed AI to assist with:

- Database development
- Backend implementation
- Frontend design
- Authentication workflows
- Machine Learning integration
- FAISS similarity search
- RCA generation
- Testing strategy
- Deployment preparation

Example Excerpt

"Develop a complete AI-Assisted Incident Management and Root Cause Analysis Platform. Design and implement authentication, incident management, severity prediction, category classification, similar incident retrieval, AI-powered RCA generation, analytics dashboards, reporting modules, and deployment documentation while maintaining production-quality architecture and security practices."

These prompts enabled rapid analysis, design iteration, implementation planning, and technical decision support throughout the project lifecycle.